

# Magic of Exponentials

## 🌾 Module 1: The ancient Indian story of the Chaturanga

### 1.1 The Legend of Chaturanga

In ancient India, the inventor of chess (Chaturanga) presented his game to the ruling king. As a reward, the inventor requested a seemingly trivial prize:

- **1 grain of rice** on the first square of the board.
- **2 grains** on the second square, **4 grains** on the third, **8 grains** on the fourth, and so on.
- The payout must double continuously for each of the **64 squares** on the board.
- The king immediately accepted, how much rice did the king owe to the inventor? Can you guess at the answer? This picture should give you a clue!



In this session we will calculate this from first principles which will show the real nature of Exponential functions following the

## 10 steps below:

1. First let us draw a 4x4 chess board with 16 squares:

|           |           |           |           |
|-----------|-----------|-----------|-----------|
|           |           |           |           |
| <b>1</b>  | <b>2</b>  | <b>3</b>  | <b>8</b>  |
| <b>5</b>  | <b>6</b>  | <b>7</b>  | <b>8</b>  |
| <b>??</b> | <b>??</b> | <b>??</b> | <b>??</b> |
| <b>??</b> | <b>??</b> | <b>??</b> | <b>??</b> |

...

2. Now, let us put the number of rice in each square:

|           |           |           |           |
|-----------|-----------|-----------|-----------|
|           |           |           |           |
| <b>1</b>  | <b>2</b>  | <b>4</b>  | <b>8</b>  |
| <b>??</b> | <b>??</b> | <b>??</b> | <b>??</b> |
| <b>??</b> | <b>??</b> | <b>??</b> | <b>??</b> |
| <b>??</b> | <b>??</b> | <b>??</b> | <b>??</b> |

...

3. This can be modeled as a series exponential numbers as follows:

[note that  $2^0 = 1$  or  $(any\ number)^0 = 1$ ]

|           |           |           |           |
|-----------|-----------|-----------|-----------|
|           |           |           |           |
| $2^0$     | $2^1$     | $2^2$     | $2^3$     |
| <b>??</b> | <b>??</b> | <b>??</b> | <b>??</b> |
| <b>??</b> | <b>??</b> | <b>??</b> | <b>??</b> |
| <b>??</b> | <b>??</b> | <b>??</b> | $2^{15}$  |

why are we missing  $2^{16}$  above ?

4. Now, let us create a Series with 16 terms for 16 squares and sum all of the terms:

$$S_{16} = 1 + 2 + 4 + 8 + 16 + 32 + 64 + \dots + 32,768$$

5. You can also write the 16-term series using exponentials as (hints:  $2^0 = 1$  and  $2^1 = 2$ ):

$$S_{16} = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^{14} + 2^{15} \text{ why are we missing the } 2^{16} \text{ term?}$$

6. Can you write out the full Series for a 8x8 square chessboard in powers of 2? (you will need 64 terms):

$$\textcircled{1} S_{64} = 2^0 + 2^1 + 2^2 + 2^3 + 2^4 + \dots + 2^{62} + 2^{63}$$

7. Multiply equation  $\textcircled{1}$  above by 2 on both sides of the = sign:

$$\textcircled{2} 2 * S_{64} = 2 * 2^0 + 2 * 2^1 + 2 * 2^2 + 2 * 2^3 + 2 * 2^4 + \dots + 2 * 2^{62} + 2 * 2^{63}$$

8. Simplify equation  $\textcircled{2}$  above to get (hints:  $2 * 2^0 = 2^{1+0} = 2^1$  and  $2 * 2^3 = 2^{1+3} = 2^4$ ):

$$\textcircled{3} 2 * S_{64} = 2^1 + 2^2 + 2^3 + 2^4 + 2^5 + \dots + 2^{63} + 2^{64}$$

9. Subtract equation  $\textcircled{1}$  from equation  $\textcircled{3}$  to get  $\textcircled{3} - \textcircled{1}$  as follows (shifted terms in equation  $\textcircled{3}$  for lining up):

|                                |          |         |         |         |         |         |            |            |          |
|--------------------------------|----------|---------|---------|---------|---------|---------|------------|------------|----------|
|                                |          |         |         |         |         |         |            |            |          |
| $\textcircled{3} 2 * S_{64} =$ | (shift)+ | $2^1 +$ | $2^2 +$ | $2^3 +$ | $2^4 +$ | $2^5 +$ | ...        | $2^{63} +$ | $2^{64}$ |
| $\textcircled{1} S_{64} =$     | $2^0 +$  | $2^1 +$ | $2^2 +$ | $2^3 +$ | $2^4 +$ | ...     | $2^{62} +$ | $2^{63}$   |          |

|                            |        |  |  |  |  |  |  |  |  |  |  |          |
|----------------------------|--------|--|--|--|--|--|--|--|--|--|--|----------|
|                            |        |  |  |  |  |  |  |  |  |  |  |          |
| $\textcircled{4} S_{64} =$ | $-1 +$ |  |  |  |  |  |  |  |  |  |  | $2^{64}$ |

10. Simplify equation  $\textcircled{4}$  above to get  $\textcircled{5}$  which represents the number of rice kernels in square 64 of the chess board.

$$\textcircled{5} S_{64} = 2^{64} - 1$$

11. Computing that that in your calculator will give you a **HUMUNGOUS** number - 18 followed by another 18 decimals or 18.5 quintillion rice kernels!!

$$\text{Total Amount of Rice} = (2^{64} - 1) = \mathbf{18,446,744,073,709,551,615} \text{ grains}$$

**How much rice is that when compared to the entire annual rice production of India? How many years will it take to produce that much rice?**

Here is some pertinent information for you to calculate this

1. Average Weight of a Single Rice Grain  $\approx$  0.025 grams
2. Indian Annual Rice Production = 150 million metric tonnes
3. There are 1000 grams in a kg
4. There are 1000 kg in a metric tonne